

## ARAXOS, Greece

WMO: 166870

Lat: 38.1492N

Lon: 21.4222E

Elev: 37

StdP: 14.68

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 33.8       | 37.1 | 22.8                          | 17.2 | 41.8 | 26.6 | 20.6 | 42.8 | 22.8                  | 52.6 | 20.6 | 53.1 | 1.0                   | 90   | 0.375 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 8             | 20.4                      | 95.2            | 71.1 | 93.1 | 70.8 | 90.0 | 71.2 | 76.9                | 86.3 | 75.5 | 85.1 | 74.5 | 84.1 | 9.9                     | 30   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 73.7                            | 125.9 | 81.7 | 73.0 | 122.7 | 81.2 | 71.5 | 116.4 | 80.1 | 40.3          | 85.9 | 39.0 | 84.5 | 37.9 | 84.8 | 86.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| 19.3              | 17.1 | 15.0 | DB                         | 29.3 | 100.2              | 2.7 | 2.7                                                | 27.4 | 102.1      | 25.8 | 103.7      | 24.3 | 105.2      | 22.4 | 107.1 | (4) |
|                   |      |      | WB                         | 27.1 | 79.5               | 2.8 | 2.2                                                | 25.1 | 81.1       | 23.4 | 82.3       | 21.8 | 83.6       | 19.8 | 85.1  | (5) |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 63.7  | 49.8  | 50.8  | 53.8  | 59.0  | 66.7  | 74.2  | 78.9  | 79.9  | 73.3  | 66.2  | 58.7  | 52.2 | (6)  |
|      |                                                                   | DBStd       | 11.52 | 4.95  | 4.95  | 4.37  | 4.57  | 4.32  | 4.82  | 3.68  | 3.72  | 4.10  | 4.46  | 5.02  | 5.12 | (7)  |
|      |                                                                   | HDD50       | 165   | 63    | 45    | 15    | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 4     | 36   | (8)  |
|      |                                                                   | HDD65       | 2066  | 470   | 398   | 346   | 187   | 32    | 1     | 0     | 1     | 39    | 194   | 398   |      | (9)  |
|      |                                                                   | CDD50       | 5165  | 58    | 67    | 134   | 271   | 517   | 727   | 894   | 928   | 698   | 502   | 266   | 103  | (10) |
|      |                                                                   | CDD65       | 1593  | 0     | 0     | 0     | 6     | 84    | 279   | 429   | 463   | 249   | 76    | 6     | 1    | (11) |
|      |                                                                   | CDH74       | 16198 | 0     | 0     | 7     | 47    | 606   | 2796  | 4957  | 5402  | 2024  | 343   | 16    | 0    | (12) |
|      |                                                                   | CDH80       | 6607  | 0     | 0     | 2     | 5     | 103   | 1052  | 2275  | 2568  | 577   | 24    | 1     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 4.7   | 5.3   | 5.8   | 5.5   | 4.8   | 4.3   | 4.0   | 4.1   | 4.4   | 4.1   | 4.7   | 4.7   | 5.2  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 25.9  | 3.5   | 3.1   | 2.4   | 1.6   | 0.8   | 0.3   | 0.2   | 0.2   | 1.1   | 3.2   | 5.3   | 5.0  | (15) |
| (16) |                                                                   | PrecMax     | 47.6  | 8.7   | 7.1   | 6.5   | 4.9   | 2.8   | 2.4   | 1.4   | 1.6   | 4.2   | 9.3   | 14.8  | 13.4 | (16) |
| (17) |                                                                   | PrecMin     | 11.5  | 0.0   | 0.2   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.2   | 0.6   | 0.8   |      | (17) |
| (18) |                                                                   | PrecStd     | 7.0   | 2.3   | 1.7   | 1.5   | 1.1   | 0.7   | 0.6   | 0.3   | 0.4   | 1.2   | 2.0   | 3.1   | 3.0  | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% DB     | 66.0  | 67.8  | 71.7  | 79.1  | 86.2  | 96.5  | 98.6  | 100.0 | 91.8  | 82.6  | 76.6  | 68.4  |      | (19) |
| (20) |                                                                   | 0.4% MCWB   | 58.4  | 58.1  | 58.6  | 62.5  | 66.7  | 71.7  | 70.6  | 71.8  | 69.2  | 68.1  | 66.0  | 61.3  |      | (20) |
| (21) |                                                                   | 2% DB       | 62.5  | 63.9  | 67.7  | 73.8  | 82.6  | 92.8  | 95.0  | 96.4  | 88.1  | 79.2  | 72.0  | 65.8  |      | (21) |
| (22) |                                                                   | 2% MCWB     | 56.4  | 56.5  | 57.8  | 61.0  | 66.4  | 71.2  | 70.8  | 70.3  | 67.0  | 63.4  | 59.6  |       |      | (22) |
| (23) |                                                                   | 5% DB       | 60.5  | 61.2  | 64.7  | 71.5  | 80.2  | 89.3  | 92.8  | 93.3  | 85.9  | 77.3  | 70.0  | 62.8  |      | (23) |
| (24) |                                                                   | 5% MCWB     | 55.3  | 54.9  | 56.7  | 59.9  | 65.9  | 70.6  | 71.3  | 70.4  | 70.7  | 66.1  | 62.6  | 57.5  |      | (24) |
| (25) |                                                                   | 10% DB      | 58.6  | 59.3  | 62.7  | 68.4  | 77.2  | 86.0  | 89.8  | 91.0  | 82.8  | 75.4  | 67.9  | 60.9  |      | (25) |
| (26) |                                                                   | 10% MCWB    | 53.5  | 53.5  | 55.8  | 58.8  | 65.4  | 70.4  | 71.7  | 71.0  | 69.8  | 65.1  | 60.7  | 56.0  |      | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% WB     | 59.9  | 59.3  | 61.7  | 65.4  | 71.8  | 76.4  | 78.9  | 78.9  | 76.8  | 72.0  | 67.3  | 62.6  |      | (27) |
| (28) |                                                                   | 0.4% MCDB   | 63.6  | 64.1  | 68.1  | 74.0  | 81.1  | 87.5  | 88.5  | 88.2  | 84.2  | 78.2  | 73.2  | 67.2  |      | (28) |
| (29) |                                                                   | 2% WB       | 57.5  | 57.3  | 59.3  | 62.9  | 69.2  | 74.6  | 77.0  | 76.8  | 74.7  | 70.0  | 65.1  | 60.2  |      | (29) |
| (30) |                                                                   | 2% MCDB     | 61.4  | 62.2  | 64.9  | 71.0  | 77.8  | 85.3  | 87.0  | 85.8  | 82.1  | 76.0  | 70.7  | 64.1  |      | (30) |
| (31) |                                                                   | 5% WB       | 55.5  | 55.7  | 57.8  | 61.2  | 67.5  | 73.1  | 75.5  | 75.5  | 73.1  | 68.6  | 63.4  | 58.5  |      | (31) |
| (32) |                                                                   | 5% MCDB     | 59.5  | 60.5  | 63.2  | 68.9  | 76.3  | 83.5  | 85.4  | 85.0  | 80.6  | 74.7  | 68.8  | 62.4  |      | (32) |
| (33) |                                                                   | 10% WB      | 53.8  | 54.4  | 56.4  | 59.9  | 66.2  | 71.8  | 74.3  | 74.4  | 71.8  | 67.2  | 61.8  | 56.6  |      | (33) |
| (34) |                                                                   | 10% MCDB    | 57.6  | 59.0  | 61.7  | 67.0  | 74.8  | 82.3  | 84.2  | 84.2  | 79.8  | 73.3  | 66.7  | 60.4  |      | (34) |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 13.9  | 14.2  | 15.7  | 17.3  | 18.8  | 20.1  | 20.7  | 20.4  | 18.2  | 16.5  | 15.5  | 14.3  |      | (35) |
| (36) |                                                                   | 5% DB MCDBR | 16.0  | 16.2  | 19.1  | 21.4  | 22.6  | 24.2  | 24.0  | 23.7  | 21.4  | 18.5  | 17.7  | 16.1  |      | (36) |
| (37) |                                                                   | 5% WB MCWBR | 11.0  | 10.5  | 11.8  | 11.2  | 10.7  | 10.0  | 8.9   | 8.6   | 9.8   | 9.6   | 10.7  | 11.1  |      | (37) |
| (38) |                                                                   | 5% WB MCDBR | 14.9  | 15.1  | 17.9  | 19.9  | 20.6  | 21.9  | 21.3  | 20.8  | 19.3  | 17.1  | 16.5  | 14.4  |      | (38) |
| (39) | Clear-Sky Solar Irradiance                                        | taub        | 0.351 | 0.381 | 0.423 | 0.464 | 0.457 | 0.436 | 0.441 | 0.451 | 0.432 | 0.409 | 0.378 | 0.358 |      | (40) |
| (41) |                                                                   | taud        | 2.437 | 2.321 | 2.194 | 2.093 | 2.175 | 2.284 | 2.274 | 2.244 | 2.292 | 2.330 | 2.383 | 2.414 |      | (41) |
| (42) |                                                                   | Ebn at Noon | 260   | 266   | 264   | 260   | 264   | 269   | 266   | 261   | 259   | 252   | 246   | 247   |      | (42) |
| (43) |                                                                   | Edh at Noon | 27    | 34    | 43    | 50    | 47    | 42    | 42    | 43    | 38    | 34    | 28    | 26    |      | (43) |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 652   | 913   | 1341  | 1768  | 2096  | 2390  | 2460  | 2192  | 1647  | 1151  | 778   | 594   |      | (44) |
| (45) |                                                                   | RadStd      | 75    | 99    | 105   | 140   | 102   | 115   | 79    | 69    | 92    | 88    | 67    | 80    |      | (45) |

## Historical Trends

|      |                         | DBAvg               | Heating           |                   | Cooling             |                   |                   | Degree-Days       |                   |                   |                   |      |
|------|-------------------------|---------------------|-------------------|-------------------|---------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|------|
|      |                         |                     | 99% DB            | 99% DP            | 1% DB               | 1% WB             | 1% DP             | HDD50             | HDD65             | CDD50             | CDD65             |      |
| (46) | Station Only            | (d)<br><b>+0.41</b> | (e)<br><b>N/A</b> | (f)<br><b>N/A</b> | (g)<br><b>+1.37</b> | (h)<br><b>N/A</b> | (i)<br><b>N/A</b> | (j)<br><b>N/A</b> | (k)<br><b>N/A</b> | (l)<br><b>N/A</b> | (m)<br><b>N/A</b> | (46) |
| (47) | Regional (10 neighbors) | <b>+0.58</b>        | <b>N/A</b>        | <b>N/A</b>        | <b>+1.39</b>        | <b>N/A</b>        | <b>N/A</b>        | <b>N/A</b>        | <b>-140</b>       | <b>+180</b>       | <b>N/A</b>        | (47) |

Nomenclature: See separate page